AI Lab-03 Task Submission

Lab03, Task01

Write a Python program that takes a string from the user and checks whether it is a palindrome or not.

A palindrome reads the same forward and backward.
Examples: madam, racecar

Hint: Reverse the string and compare it with the original string.

Lab03, Task02

Given a list of integers, write a program to calculate the average of all elements in the list.

Lab03, Task03

Take two lists of integers as input from the user. Merge the lists into one list and sort the resulting list.

Lab03, Task04

Given a tuple of integers, create:

- A tuple containing all even numbers
- A tuple containing all odd numbers

Lab03, Task05

Create a dictionary where:

- **Keys:** Student names
- **Values:** Marks (integers)

Create a menu-based program with the following operations:

A — Add a student
B — Update student marks
C — Search for a student
D — Display all students and marks

Lab03, Task06

Given a list of words, create a dictionary that maps each word to its length.

Example:
"Python" → 6

Lab03, Task07

Write a Python program that takes a string from the user and counts the number of spaces in the string.

Lab03, Task08

Write a program to check whether two lists have no common elements.

Hint: Use sets.

Lab03, Task09

Given a list, write a program to display all elements that appear more than once.

Hint: Use sets.

Lab03, Task10

Write a Python program that takes a string from the user and displays:

- All unique characters
- The number of unique characters